Aryan B. Patel

Calhoun, Georgia | 706-263-5996 | abp.aryan7@gmail.com |abp-aryan7.github.io | www.linkedin.com/in/abp-aryan7 | US Citizen

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
Bachelor of Science in Electrical Engineering - Rob/Aut Sys & Sens/Explor **Expected Graduation, May 2029**
Bachelor of Science in Mechanical Engineering **GPA 4.00**

- Relevant Coursework: Multivariable Calculus, Differential Equations, Linear Algebra, Physics I & II, Statics, Engineering Materials, Circuit Analysis, Digital System Design, Computing Techniques

Skills

Programming: C/C++, Python, Java, JavaScript, MATLAB, Numerical Methods
CAD & Engineering: SolidWorks, 3D Printing, Rapid Prototyping, Engineering Drawings, Optical System Design, Beam Alignment
Digital Logic and Hardware: Digital Logic Design, Computer Architecture Fundamentals, Datasheet Analysis, IC-Level Debugging, Breadboarding, Solder Iron, Multimeter
Embedded Systems: Arduino, Raspberry Pi, Stepper Motors, Motor Drivers, Microstepping, Interrupt Handling
Tools: Git, GitHub
Languages: English (Fluent), Gujarati (Fluent), Hindi (Conversational), Japanese (Beginner)

Projects

Laser Projector | Personal Project *April 2026-August 2026*
Individually designed and researched a compact laser projection system focused on portability, visibility, and cost efficiency.

- Designed and built a two-axis laser projection system using a rotating polygon mirror and stepper-controlled mirror for horizontal and vertical beam scanning.
- Programmed an Arduino Nano to synchronize motor motion and laser timing, generating 12-line raster patterns, bitmap graphics, and scrolling text.
- Designed and iterated custom 3D-printed mechanical components for motor, mirror, and optical alignment through prototype testing.

Solar System Simulation | Personal Project *December 2025-April 2026*
Developed an interactive solar system simulation to help model planetary movements and orbital mechanics through real-world physics as an individual project.

- Independently developed an interactive solar system simulation modeling planetary motion and orbital mechanics using real-world physics.
- Researched and implemented advanced simulation methods to model nearly 1.2 million asteroids and 5,000 comets.
- Designed simulation algorithms to balance computational performance with physical accuracy across large numbers of celestial objects.

8-Bit Breadboard Computer | Personal Project *September 2024–November 2024*
Independently built a complete Ben Eater-architecture 8-bit computer from discrete logic ICs, researching and substituting components unavailable within budget constraints.

- Constructed the full system from breadboarded logic circuits (clock, registers, ALU, control logic, and CPU) based on Ben Eater's architecture.
- Researched and cross-referenced datasheets to identify pin-compatible, functionally equivalent IC substitutes when original components were unavailable or cost-prohibitive.
- Diagnosed and resolved signal-level integration issues to get substituted chips working correctly within the existing control logic and timing.
- Programmed and ran custom counting sequences, including a skip-by-twos program, demonstrating functional control logic and instruction execution.

Experience

BAPS Community | Calhoun, Georgia *December 2020– Present*
Volunteer

- Led coordination of multiple youth events with 200 participants.
- Organized weekly youth activities and engagement events for 5 years.
- Assisted in choreographing and coordinating a traditional Indian dance involving 20+ youth participants.

UPS Store | Calhoun, Georgia *June 2023– Present*
Store Assistant

- Helped customers with shipping, packing, printing, and mailbox services.
- Developed communication, problem-solving, and multitasking skills through customer interactions.
- Collaborated with staff to maintain efficient store operations in a fast-paced environment.